

Nonnegative and positive factorizations

Polynomial-time minimum-volume decision under sufficient scattering

Difficulty: extreme **Importance:** broadly interesting

Topic: structured matrix factorization; global optimization

Status: open; admitted after a bounded literature search found no resolution.

Context and notation

For $H \in \mathbb{R}_+^{r \times n}$, let $\text{cone}(H) = \{Hy : y \in \mathbb{R}_+^n\}$ and let e_d be the all-ones vector in $\mathbb{R}^d$. Here **SSC** means the precise version in Gillis's Definition 4.15:

$$C_r = \{x \in \mathbb{R}_+^r : e_r^T x \geq \sqrt{r-1} \|x\|_2\} \subseteq \text{cone}(H),$$

and every orthogonal $Q \in \mathbb{R}^{r \times r}$ satisfying $\text{cone}(H) \subseteq \text{cone}(Q)$ is a permutation matrix. Some later sources use a stronger second condition involving the boundary of the dual cone; that condition is not substituted here.

For $X \in \mathbb{R}^{m \times n}$ and $r = \text{rank}(X)$, the minimum-volume problem considered here is

$$\min_{W \in \mathbb{R}^{m \times r}, H \in \mathbb{R}_+^{r \times n}} \det(W^T W) \quad \text{subject to} \quad X = WH, \quad e_r^T H = e_n^T. \quad (\text{MV})$$

In particular, W is unrestricted in sign and the **columns** of H sum to one. This is min-vol NMF (1), Definition 4.42 of the book.

Problem statement

Consider the following rational-input decision version of minimum-volume optimization. The input is $X \in \mathbb{Q}^{m \times n}$, an integer $2 \leq r \leq \min(m, n)$, and $\tau \in \mathbb{Q}_{\geq 0}$, with the promise that $\text{rank}(X) = r$ and that there exist real factors $X = W_* H_*$ feasible for (MV) with H_* satisfying SSC. These factors are not supplied. Decide whether

$$\exists W \in \mathbb{R}^{m \times r}, H \in \mathbb{R}_+^{r \times n} : X = WH, \quad e_r^T H = e_n^T, \quad \det(W^T W) \leq \tau.$$

Does a polynomial-time algorithm exist, with time measured in the total binary input length? Randomization is allowed with success probability at least 2/3 on every promised input. The algorithm must run in polynomial time on all inputs but need only answer correctly on promised inputs. Checking the SSC promise is not part of the task. This decision formulation fixes the computational model of the book's polynomial-solvability conjecture; it does not assert an equivalence with exact factor recovery.

References

Gillis, *Nonnegative Matrix Factorization*, §4.3.3.6, pp. 148–149, with Definitions 4.15 and 4.42 and Theorem 4.43; author-hosted book. Barbarino, Gillis, and Saha, *Robustness of Minimum-Volume Nonnegative Matrix Factorization under an Expanded Sufficiently Scattered Condition*, §5, final research question.

Status check

Searches for *minimum-volume NMF polynomial time sufficiently scattered* and *minimum volume simplex sufficiently scattered algorithm* 2026 2025 found no polynomial-time guarantee for this promise. The November 2025 paper still asks for complexity results under its stronger p -SSC assumption. Its robustness theorems assume a globally optimal minimum-volume solution; they do not compute one in polynomial time. The book explains why the maximum-inscribed-ellipsoid approach can require exponentially many polytope facets.